

Technology Stack

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Project Name	Credit Card Approval Prediction System
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Define Your Technology Stack

The technologies chosen to build the Credit Card Approval Prediction system, reflecting what is currently implemented in the project repository, with recommended additions noted where a component is not yet in place.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML/CSS with Jinja2 templates (index.html, result.html) rendered by Flask	The interface is a single input form and a result page - no complex client-side state is needed, so server-rendered HTML keeps things simple and fast to build without a JS framework.
2	Backend / Server-Side	Python + Flask	Flask is lightweight and integrates directly with the same Python/pandas/scikit-learn stack used to train the model, so the trained model.pkl can be loaded and served with minimal glue code.
3	Database / Data Storage	Flat CSV files (application_record.csv, credit_record.csv), read with pandas - no live database is used at runtime	Sufficient for a batch-trained offline model where training data doesn't change per request. Recommended upgrade: PostgreSQL or MySQL to persist submitted applications and predictions if the app needs to track applicant history.
4	Cloud / Hosting / Deployment	None yet - currently runs locally via Flask's development server (app.run(debug=True))	Fine for local development and demos. For production, recommend deploying via Render, Heroku, or a Docker container on AWS/Azure behind a WSGI server such as Gunicorn.
5	Version Control & CI/CD	GitHub for source control; no CI/CD pipeline configured yet	GitHub gives free hosting, version history, and easy sharing for a student/portfolio project. A future addition of GitHub Actions could automate linting/tests on each push.
6	Third-Party APIs / Other Tools	scikit-learn (LogisticRegression, DecisionTreeClassifier, RandomForestClassifier) and pickle for model persistence; no external/third-party API is called at runtime	scikit-learn provides ready-to-use, well-tested ML algorithms so the team can compare multiple models quickly; pickle lets the trained Random Forest model be reused instantly by the Flask app without retraining on every request.